

PHOONG GUIK MING

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Portfolio: <https://pgm789.github.io/portfolio/>

Availability: Immediately

SUMMARY

Computer Science recent graduate with hands-on experience across backend development, IoT data analytics, and cloud computing. Proficient in building web applications, REST APIs, and automated data pipelines using Python, Flask, and AWS. Eager to leverage backend and data analytics skills to develop, test, and optimize scalable tech solutions.

TECHNICAL SKILLS

- **Programming Skills:** C++, Java, Python, R, SQL Plus, HTML, JavaScript, PHP, Linux
- **Tools:** Visual Studio Code, Visual Studio 2022, Jupyter, Colab, Pycharm, Eclipse, Streamlit, XAMPP, AWS, Oracle VM Virtual Box, VMware, Cisco Packet Tracer, GNS3, Postman, Microsoft (Excel, Word, Power Point, Access)

INTERNSHIP EXPERIENCE

Sophic Automation Sdn Bhd

Oct 2024 - Jan 2025

Software Innovation Department

- **Automated Data Scraping:** Built web scrapers to automate data collection, replacing manual gathering workflows.
- **Full-Stack API Development:** Developed web applications and built Flask REST APIs to handle data flow and connect different parts of the internal systems.
- **Pumphouse Water Level & Traffic Analytics Forecasting:** Deployed ARIMA and VAR time-series models to analyze live pumphouse water level data for trash clearance, and adapted the framework to toll plaza datasets to forecast peak hours and pinpoint high-volume booths.
- **Computer Vision Systems:** Implemented object detection and tracking models to automate visual monitoring and data collection tasks.

FINAL YEAR PROJECT

Feb 2025 - Sep 2025

Real-Time Attendance System (Raspberry Pi 5, Ultrasonic sensor, PIR sensor, Python, OpenCV, Dlib, MySQL, HTML, Debian Linux):

- **Multi-Sensor Trigger System:** Designed an energy-efficient hardware pipeline using a Raspberry Pi 5 integrated with a PIR motion sensor and an ultrasonic distance sensor to automatically wake the system and trigger the webcam when a user is within 50cm.
- **Computer Vision & Liveness Detection:** Developed a robust facial recognition pipeline using OpenCV and Dlib to extract frames, track faces with color-coded bounding boxes, and implement an eye-blink detection step to prevent photo spoofing.
- **IoT Data & API Integration:** Built a Python-based backend that updates student attendance records in real time via a Flask REST API and logs data to a structured database.
- **Web Interface & Automated Alerts:** Designed an HTML web dashboard to display live attendance tables and configured automated low-attendance warnings (under 80%) pushed instantly via a Telegram bot.

ACADEMIC PROJECTS

- **Movie Review Web (HTML, JavaScript, PHP, MySQL, Visual Studio Code):** Created a web application with user authentication and CRUD functionality for movie reviews. (<https://github.com/pgm789/moviereviewssystem.git>)
- **UniteFund Mobile App (Java, Android Studio):** Developed a crowdfunding platform where users can raise money for various personal and charitable causes. (<https://github.com/pgm789/G19-NickGurr.git>)
- **Traffic Sign Detection System (C++, OpenCV, Visual Studio 2022):** Developed a traffic sign detection system using image preprocessing, color segmentation, contour detection, and object recognition.
- **Diagnosis and prevention of cardiovascular disease (Python, Jupyter):** Train machine learning models that can accurately predict the cardiovascular risk (low, medium, high) of an individual.
- **Real-Time Weather & Flood Risk Pipeline (AWS):** Built a cloud-based pipeline for processing weather data and flood risk information.
- **Stock Management System (Java, Object-Oriented Programming, Eclipse):** Developed a Java-based stock management system using object-oriented programming principles to manage product information and inventory operations.
- **Point of Sales (POS) System (SQL, Microsoft Access):** Designed a relational database for food ordering, sales and restaurant operations.
- **Augmented Reality (AR) Interactive Object Viewer (C#, AR Foundation, Unity):** Developed an AR interactive object viewer that enables users to rotate 3D objects, display object information, and play audio descriptions upon image marker recognition. (<https://github.com/pgm789/AR1.git>)
- **Virtual Reality (VR) Treasure Hunt Game (C#, XR Interaction Toolkit, Unity):** Developed a VR treasure hunt game featuring teleportation-based navigation, interactive object collection, and immersive gameplay using Unity's XR framework. (<https://github.com/DevilGemini/New-VR-Hunter.git>)
- **Network Design (Cisco Packet Tracer, GNS3):** Designed and simulated network topologies with routing and VLAN configuration.
- **Cybersecurity Virtual Lab (VirtualBox, Linux):** Created a virtual environment for password auditing and cybersecurity experiments.

EDUCATION

University Tunku Abdul Rahman (UTAR KAMPAR CAMPUS)

Oct 2020 - June 2026

- Bachelor of Computer Science (Honours)

ADDITIONAL INFORMATION

- **Languages:** English, Malay, Mandarin, Hokkien
- **Certifications:** AWS Academy Cloud Foundations, AI in Application Design(ASEM)
- **Extra Curricular Activities:**
 1. **Chairperson, UTAR Chinese Buddhist Talk (Feb 2024)** – Led committee and event execution.
 2. **Publicity Manager, Meditation Camp (Jan 2024)** – Handled poster design, social media marketing, photography, and video editing.
 3. **Technical Manager, UTAR Welcoming Night (Jan 2023)** – Handled wiring and troubleshooting of projectors and audio systems.
 4. **Logistics Assistant, Annual Buddhist Camp (Jan 2023)** – Arranged venue, catering, and transportation logistics.
 5. **Program Assistant, Annual Buddhist Camp (Jan 2024)** - Managed event timeline, speaker relations, and cross-team coordination.
 6. **Volunteer, UTAR Hospital (Mar 2026)** – Contributed 28 hours in patient navigation, wheel-chair assistance, and registration.